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Radiation tolerant frequency comb fiber laser for space applications

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ABSTRACT

Advanced radiation tolerant Erbium doped fibers have been developed surviving an accumulated dose of 1 kGy*. Femtosecond fiber lasers and amplifiers manufactured from such fibers have been packaged and irradiated under active operation to test for accelerated ageing with dose rates up to 10 mGy/s using a Cobalt 60 source. The laser output power degraded by about 11% after exposure of a dose of 1 kGy. Compared to conventional Erbium fiber lasers this represents a significant improvement in radiation tolerance.

Keywords: frequency comb, radiation hard, erbium doped fiber laser

1. INTRODUCTION

Since its invention in 1999 the optical frequency comb (OFC) has become the Swiss-knife to provide and measure optical frequencies, unavoidable for advanced optical clocks and precision quantum sensing.¹ Today, there exists a growing demand for a space qualified OFC, covering many applications ranging from precision atomic clocks, ranging, LIDAR, gravitational sensing, and specific fundamental physics missions based on ultra-precise measurements. Since 2010 Menlo Systems has advanced its comb technology, aiming for reduced size, weight and power (SWAP), and has hardened these systems against environmental effects. Several sounding rocket missions of compact, low power and automated combs have been performed, demonstrating the feasibility of precision spectroscopy in space.^{2,3} Since the beginning of our activities the radiation hardness of the doped fibers involved has been of major concern. The radiation sensitivity of optical fibers is usually classified into three different types:⁴ (a) radiation sensitive, (b) radiation tolerant, and (c) radiation hardened. Fiber lasers based on rare earth doped fibers using Yb and Er are known to be radiation sensitive. A recent paper reviews the fiber degradation caused by energetic particles.⁵ Radiation degradation of fibers is complex and depends on many parameters. Intrinsic ones are mainly related to fiber chemical composition and fiber guiding properties, but also extrinsic parameters exist, like the type of radiation, the accumulated dose, the dose rate, the temperature, the operating wavelength, and the optical power in the fiber core.

Energetic radiation leads to the generation of point defects in the fiber core, increasing the absorption at pump and lasing wavelengths. However, many point defects appear to be unstable and decay or can be annealed by temperature or light absorption. As a consequence, the ratio of the dose rate of energetic radiation in combination with the laser operation at its nominal power influence the forward-backward reaction between the generation and the annealing of defects. Higher core temperature favors faster annealing, and an increased optical pump power in the fiber has a similar effect. This is of particular importance to active fibers pumped in the visible or near infrared wavelength range, as various color centers exist in these wavelength regions. Therefore we have studied our fiber lasers under active operation using varying on-off duty cycles.

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*The accumulated dose D is expressed in Gray, 1 Gy corresponds to absorption of 1 J/kg, or 100 Rad. The dose rate dD/dt is the dose absorbed per time unit and expressed in Gy/s

Er doped fibers are at particular risk to radiation damage⁶ in fiber oscillators. Energetic radiation causes radiation induced absorption (RIA) at both the pump and the signal wavelength.⁷ This results in decreased amplification efficiency nearly independent of the type of the ionizing particle.⁸ An empirical model allows to estimate the RIA dependence from a steady-state radiation,⁹ it can reasonably be reproduced by a power law function $A(D) = C_0 D^\beta$, where $A(D)$ is the RIA, D is the accumulated dose, and β and C_0 are constants depending on fiber properties and operating wavelength.

2. EXPERIMENTAL

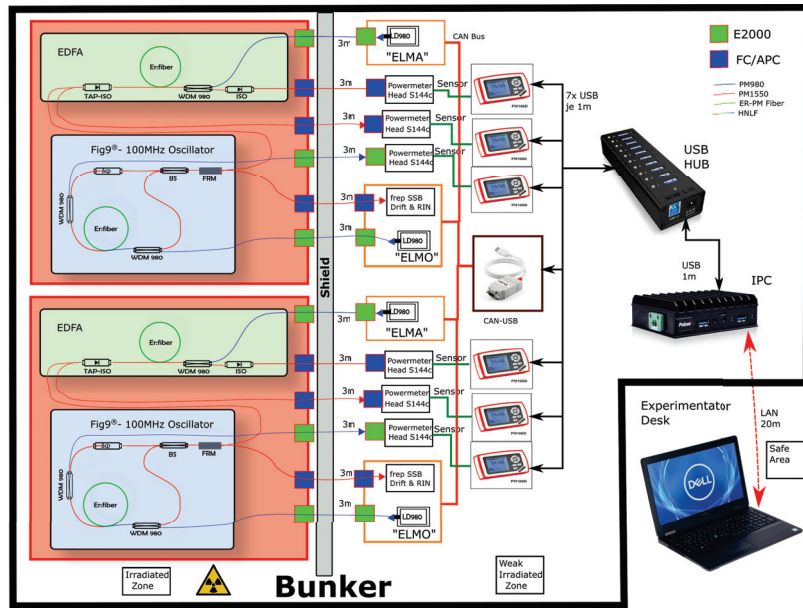


Figure 1. Irradiation set-up used for the characterization of radiation hardness of fiber lasers and amplifiers under active operation. Two laser oscillators and amplifiers were characterized in parallel. Electronics and diagnostics have been shielded from radiation and the active set-up was fully controlled from outside the irradiation bunker.

In the past years iXblue has developed and commercialized radiation tolerant rare-earth doped fibers. A comprehensive overview on the present state of the art of design, simulation, and testing of radiation hard fibers for space applications has been provided by Ladaci.¹⁰ Based on such work it was possible to develop fast-track an advanced gain fiber matching a goal that has been twofold: (a) reduced radiation sensitivity at an increased dopant concentration and (b) matching the fiber dispersion to fiber laser oscillator requirements. Besides reduction of RIA which reduces also the radiation induced gain variation (RIGV), we have also studied here the influence of laser activity on the accelerated ageing.

Several femtosecond fiber oscillators¹¹ and amplifiers have been manufactured for irradiation experiments. The fiber layout has been quite similar to the fiber layout used in previous OFC sounding rocket campaigns,³ but it contains an additional wavelength divider to study of RIA in an active oscillator. On the other side, the test lasers have been manufactured without the actors necessary for the stabilization of the two fundamental frequencies of the OFC. Omitting such actors, which are basically electro-optic modulators and piezo stacks, has very limited influence on the lasing operation itself, while the laser manufacturing time can be significantly reduced. However, it is foreseen in the project that for the final demonstration at ESA a complete fiber comb design with all required actors will be manufactured and irradiated.

Fiber oscillators and amplifiers have been irradiated at INT in parallel by a calibrated Co-60 Gamma ray source, providing photons at 1.172 MeV and 1.332 MeV. Fiber lasers have been symmetrically positioned at about 20 cm from the source to guarantee identical irradiation of 10 mGy/s. Both, pump and diagnostic light passed through 3 m long buffered patchcords to the laser driver units and the diagnostic power meters. Laser

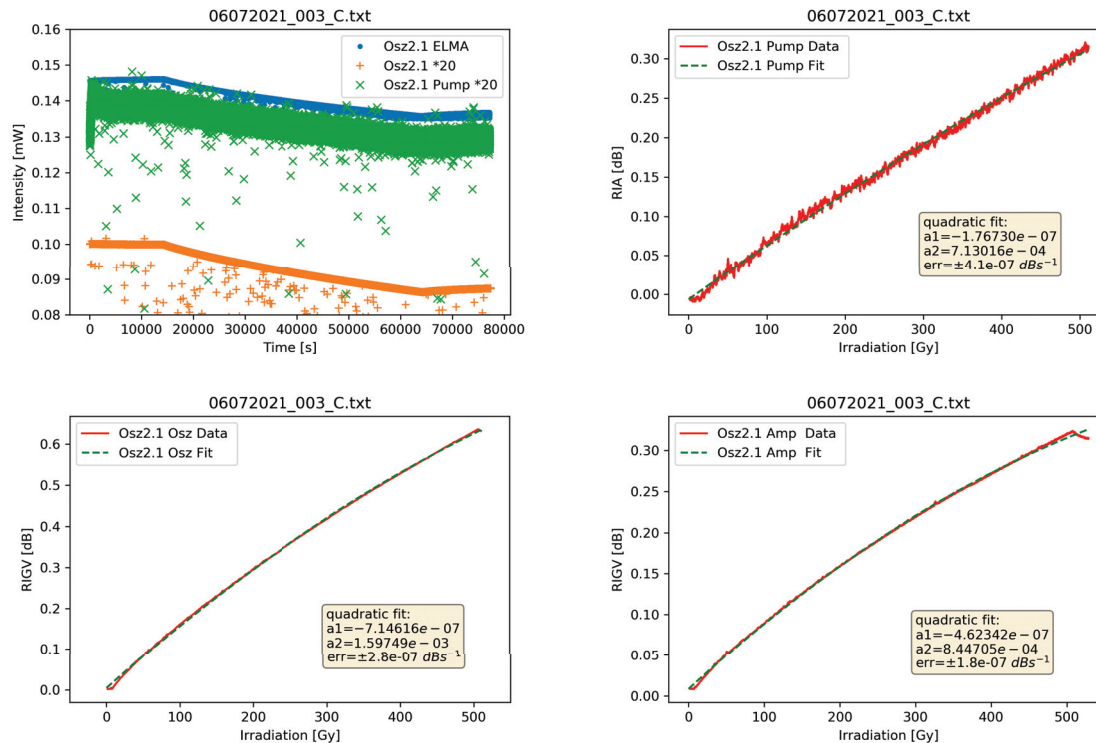


Figure 2. Typical data collected at INT for a laser under test during 500 Gy irradiation: Upper left shows raw data from power meters at different inspection points. Upper right shows the derived increasing pump light absorption in the oscillator (RIA) with scattered points caused by on-off switching of the lasers. Lower left shows the degradation of oscillator output and lower right the same for the power amplifier output seeded by the oscillator (RIGV). The raw data has been fit by quadratic functions, as indicated.

operation and power meters have been controlled and read out by an industry PC. While laser oscillators and amplifiers have been subject to intense Gamma irradiation, all electronics was efficiently shielded against any radiation damage. The complete experimental set-up, shown in figure 2, was remotely controlled from outside the irradiation bunker. At the selected dose rate of 10 mGy/s the accumulation of 1 kGy is achieved within roughly 27.8 hrs. The irradiation duration has been split into two 13.9 hrs phases, with a few hrs of recovery and inspection in between, enabling also the observation of RIA and RIGV recovery rates of the irradiated lasers. Oscillator and amplifier have been operated at both, 10:90 and 90:10 duty cycle, to monitor annealing effects caused by the amount of active laser operation. A typical experimental data-set of such experiments is shown in figure 2.

3. RESULTS AND CONCLUSION

Table 1. Accelerated ageing for a radiation tolerant compared to a standard fiber laser, measured at different duty cycles, and for radiation induced absorption (RIA) and radiation induced gain variation (RIGV).

	Oszilator RIA [$\text{dB} \cdot \text{Gy}^{-1} \cdot \text{m}^{-1}$]		Amplifier RIGV [$\text{dB} \cdot \text{Gy}^{-1} \cdot \text{m}^{-1}$]	
Duty Cycle	Standard	Radiation Tolerant	Standard	Radiation Tolerant
10%	6,94E-03	4,66E-03	8,23E-03	1,87E-03
90%	6,16E-03	1,59E-03	5,94E-03	8,44E-04

Both, RIA and RIGV have been found to depend on the duty cycle and the pump light intensity, see table 1.

It becomes clear that the survival of a fiber laser in space is counter intuitively significantly increased when the laser will be operated over extended periods. Because precision measurements usually require long term comb operation to obtain sufficient statistics, this is an advantageous behaviour of the underlying ageing process. Compared to standard fibers, the fiber ageing is found to be significantly decreased by a factor 0,16. Moreover, we did not observe here any significant degradation of other fiber laser components than the gain fiber. The loss of output power after 1 kGy has been 11% for the oscillator and amplifier at 90% duty cycle and 1 kGy, compared to about 25% for the standard gain fiber irradiated with 200 Gy for comparison. The remaining performance loss of our radiation tolerant laser technology could additionally be compensated by: (a) About 1:3000 reduced irradiation in space at MEO compared to the present study which will shift the forward-backward reaction largely towards an increased lifetime; (b) continuous adjustments of pump power during the mission can be applied to compensate for radiation induced loss of laser power. From the present data we therefore conclude that laser systems build from the components used in the present experiment should survive at least 10 yrs at Mid-Earth Orbit (MEO), provided that they are additionally shielded by about 6 mm Aluminum.

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